

Critical Paper Analysis: Solving Inequality Proofs with Large Language Models

- Authors: Pan Lu, Jiayi Sheng, Luna Lyu, Jikai Jin, Tony Xia, Alex Gu, James Zou (Stanford, Berkeley, MIT)
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- Presented by: BOULGHERAIF Fatma Zohra
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Paper at a Glance (1/2)

Contributions

1. IneqMath dataset: 200 test + 1,252 training problems
2. Informal reformulation: Bound estimation + Relation prediction
3. LLM-as-judge framework: 1 final-answer + 4 step-wise judges
4. Evaluation of 29 leading LLMs

Resources

Dataset, code, and leaderboard publicly available

Paper at a Glance (2/2)

Key Finding

- "Even top LLMs like o1 achieve $<10\%$ overall accuracy on inequality proofs when step-wise reasoning is scrutinized"
- Answer Accuracy vs Overall Accuracy drops up to 65.5% → Finding answers $\neq$ constructing rigorous proofs

Task & Problem Formulation (1/2)

Task Formulation: “Informal Yet Verifiable”

- Split into 2 subtasks:
 - Subtask 1: **Bound Estimation**
Find maximal C such that $f(x) \geq C \cdot g(x)$ holds
 - Subtask 2: **Relation Prediction**
Determine relation between $f(x)$ and $g(x)$
- + Claimed benefit: preserves reasoning challenges
- Assumption: All inequality proofs fit this format

Task & Problem Formulation (2/2)

Is this reformulation valid?

Strengths	Concerns
• Clear mathematical formulation. • Final answers automatically verifiable. • Avoids formal proof assistant complexity.	• No empirical proof that reformulation maintains difficulty. • Do inequality proofs naturally fit? • “Verifiable” overstated: final-answer: yes, step-wise: black-box.

Dataset Construction

Scale:

Split	Train	Dev	Test
Total	1252	100	200

Quality >>> Quantity

Dataset Construction

Curation process:

Test Set (200 problems)

- Novel problems created by IMO medalists
- Rigorous expert review panel
- Validated for solvability & correctness

Training Set (1,252 problems)

- Sourced from advanced textbooks
- Auto-rephrased by LLMs → Human validation
- 76.8% annotated with 83 named theorems
- Up to 4 step-wise solutions per problem

Benchmark Design

- Evaluating the performance of 29 LLMs (open source and proprietary | general purpose and reasoning) on ineqmath
- No comparison with baselines (proof assistant solvers)

Benchmark Design

- **LLM-as-Judge Evaluation Framework**

- Final Answer Judge
- 4 Step-wise Judges:
 - Toy Case Judge
 - Logical Gap Judge
 - Approximation Judge
 - Computation Judge

- **Metrics**

- Answer Acc: Final answer correct?
- Step Acc: Each reasoning step sound?
- Overall Acc: Primary metric

Experimental Protocol

- **Reproducibility and clarity:** Good reproducibility for dataset, moderate for exact evaluation replication (LLM-as-judge prompts are deterministic but models change)
- **Hyperparameter fairness:** All models evaluated zero-shot.
- No statistical tests

Strengths

1- Novel Evaluation Paradigm

- separate answer correctness from proof soundness
- step-wise judges targeting specific error types
- reusable framework for other domains

2- High-Quality Expert Curation

3- Comprehensive LLM Evaluation

- 29 models tested systematically
- reveals important gap in reasoning capabilities

4- Useful Community Resource

5- Important Insight

- challenges assumption that answer accuracy = understanding
- highlights need for rigorous reasoning evaluation

Weaknesses & Limitations

- LLM-as-Judge Reliability
- Task Validity not Proven
- Missing Key Baselines
- Limited Statistical Rigor
- Scale Limitations

Comparison to Prior Work

Moderately incremental

- **Novel:** Combination + step-wise LLM judges
- **Not novel:** Informal math, inequality focus, Olympiad problems

Each component exists separately; **innovation** is the **combination**

Ethics & Responsible Use: Not a major concern

OpenReview Reviews – Summary

Score distribution: average **4.8/6**

Reviewer	Score /6	Confidence /5	Final Decision
CZwA	5	4	Accept
j6VP	5	3	Accept
FVP8	4	3	Borderline -> 5
euVH	6	4	Strong Accept
2Aps	4	5	Borderline

OpenReview Reviews – Summary

1- Data Scale [raised by: FVP8, j6VP]

Concern: "1,252 samples too small for D&B track"

Expected: 5K-12K samples

Impact: Limits statistical power, training

2- Automation Limitations [raised by: J6VP]

Concern: "Heavy expert involvement limits scalability"

Impact: Cannot easily expand dataset

OpenReview Reviews – Summary

3- Difficulty Stratification [raised by: 2Aps]

Concern: "Unclear problem difficulty distribution"

Impact: Hard to assess performance bt levels

4- LLM-as-Judge Validation [raised by: FVP8, CZwA]

Concern: "Only 80-100 dev samples validated"

"No cross-judge agreement study"

"Systematic biases not analyzed"

Impact: Can we trust the evaluation?

OpenReview Reviews – Summary

5- Contamination Analysis [raised by: FVP8]

Concern: “Weak analysis of data leakage”

“No memorization probes”

Impact: Results might reflect memorization

6- No Formal Prover Comparison [raised by: FVP8]

Concern: “Missing Lean/Isabelle baselines”

Impact: Can't compare informal vs formal

Rebuttal Analysis – Successfully Addressed

- **Dataset scale -> resolved**

1252 -> 5k, generated in <1 hour using automated pipeline
demonstrated scalability of approach, reviewers satisfied

- **Difficulty stratification -> resolved**

proposed then implemented multi-criteria classification,
added to dataset
reviewer satisfied

Rebuttal Analysis – Successfully Addressed

- **Contamination analysis -> addressed**

Added memorization probe. Test problems hand-crafted
reviewer accepted explanation

- **LLM-as-Judge validation -> partially addressed**

Additional 100 test samples manually labeled, added bias
analysis

Reviewer: Still some skepticism but improved

Rebuttal Analysis – Poorly Addressed

- **LLM-as-Judge Fundamental Concerns**

What reviewers wanted:

- Validation on 500+ samples (not 100)
- Cross-judge agreement study
- Position bias analysis
- Comparison with human expert evaluation

The response was surface-level, evaluation still black-box, and core reliability concerns were not addressed

Rebuttal Analysis – Poorly Addressed

- **Formal Proof Solver Comparison**

The initial response (“Different Paradigm”) was dismissive and evasive

Only added results under pressure

- **Statistical Rigor**

Critical Reasoning Exercise

- If I were a reviewer: accept
- Suggestions:
 - in-depth analysis of llm performance, the failure analysis is very shallow.
 - contextualize llm performance by comparing to human performance.
 - explain inequality proofs better, and why dividing to the subtasks is a good fit (maybe it is my lack of knowledge on the subject)